Dissertation review: decisions and comments

31 October 2026 | Review only: no text replaced

First decision: abstract

A: restore the exact latest downloaded Overleaf version. B: retain the current local version. C: retain the local version with first-person BPA characterization. No selection has been applied.

A. Downloaded Overleaf abstract

Biomimetic robots apply biological knowledge to robotics and enable controlled experiments in biomechanics and neuroscience. This dissertation addresses four needs for a human-proportioned robot with braided pneumatic actuators (BPAs): force prediction across resting lengths, corrected joint-torque predictions, accessible sensory-afferent evidence, and tools for transferring and testing neural controllers between biomechanical models. Festo BPA characterization across resting lengths, contractions, and pressures produced a resting-length-dependent maximum-force model and a three-coefficient nondimensional force surface. Joint-torque design factors account for length offsets, effective series compliance, and wrapping-induced loss of usable actuator length. Redesigned paired BPA flexor and extensor routes met or exceeded selected human-muscle reference torques at the measured knee angles. The Sensory Afferent Database (SADb) connects experimental findings and model assumptions through searchable notes and citation navigation. Neural-model tools, graphical blocks, and simulation interfaces enable controlled experiments in MuJoCo and Simulink/Simscape. Evaluations establish selected circuit responses, implementation agreement, and virtual-harness-assisted balance and walking behaviors. Together, these contributions support experimentally informed robot design and neuromechanical experiments, with potential prosthetic and orthotic applications.

B. Current local abstract

Musculoskeletal humanoid robots enable controlled experiments in biological movement and motivate compliant locomotion in unstructured environments. Their development requires reliable actuator and joint models, physiological evidence, and tools for testing neural controllers with different mechanics. I address these needs through four connected contributions. First, Festo braided pneumatic actuator (BPA) characterization across resting lengths, contractions, and pressures produced a resting-length-dependent maximum-force model and a three-coefficient nondimensional force surface. Joint-torque design factors then account for length offsets, effective series compliance, and wrapping-induced loss of usable actuator length. Incorporating these factors into route redesign enabled paired BPA knee flexor and extensor routes to meet or exceed selected human-muscle reference torques at measured configurations. I created the Sensory Afferent Database to connect experimental findings and model assumptions through searchable notes and citation navigation. Graphical neural blocks and simulation interfaces support controller transfer and testing in MuJoCo and Simulink/Simscape. Representative evaluations establish circuit responses, implementation agreement, and virtual-harness-assisted balance and walking behaviors. These contributions connect experimentally informed robot design with neuromechanical experiments and potential prosthetic and orthotic applications.

C. Minimal proposed change

Replace only the characterization sentence in B with: First, I characterized Festo braided pneumatic actuators (BPAs) across resting lengths, contractions, and pressures and developed a resting-length-dependent maximum-force model and a three-coefficient nondimensional force surface.

Background choice

Restore the latest downloaded background, or keep the local version. They differ only at source line 102: "corrected torque model" in the download versus "improved torque model" locally. The downloaded version already contains earlier background revisions; it is not a pre-revision original.

Fetched evidence: corrected status

1. Circuit drawings ARE present locally

Incoming W2L source-network drawings are in Neuromechanical_Models/W2L_VirtualWalker_Newer_20261031/subnetwork_review, including exact_LH.svg, exact_LH_RG.svg, and PF/MN/commissural companions. These are distinct from the two rejected proposals. The open task is to select the appropriate drawing, not to create it again.

2. New recording evidence IS documented in the fetch

CANONICAL108_STIMULUS_GRAPHS.md identifies actual MuJoCo W2L baseline and RG/PF/MN pulse recordings with provenance. The saved images are under easteregg2 runs folders and may not be tracked by Git. These suspended MuJoCo trials must remain distinct from native AnimatLab and platform-drop recordings. Handoff prompts below request exact files and trial identities.

3. Routing constraint and future work

Ben confirms the search was instructed not to increase musculotendon length further. Retained attachment locations are consistent with that constrained design task. Changing cost-function priorities belongs in future work. No optimization, fit, or Table 4.12 number changes are needed.

4. Remaining content choices

Select the abstract/background baseline; select the fetched source-circuit drawing; identify the precise native/drop recording if it is to be shown. Leo S05/S09 still lack historical manual-search examples and a genuine AI-triage worked example. Existing monthly-watch terms and personal speed/quality judgment are already described.

All Leo messages follow

58 messages across 57 threads, through 30 October 9:07 pm Pacific. Each card shows the captured comment followed by the current disposition. Existing live comments have not been resolved or answered.

A01 - 30 October 2026, 6:12 pm Pacific

There's no real set up for this sentence. You mention biomimetic robotics in general and then jump to human robots

Addressed: connected one-page abstract; first-person ownership retained. | 02-abstract.tex:7

A02 - 30 October 2026, 6:14 pm Pacific

It would be nice if there was a logical flow to these points or a story you're telling with the project rather than just a list

Addressed: connected one-page abstract; first-person ownership retained. | 02-abstract.tex:7

A03 - 30 October 2026, 6:15 pm Pacific

This sentence would be easier to understand if it was in the first person. 'I produced a x model by...'

Author review under freeze: the actuator-characterization sentence remains passive; no further abstract edit. | 02-abstract.tex:7

A04 - 30 October 2026, 6:19 pm Pacific

I think each of these are effectively chapter conclusion summaries but they need some more set up. A thesis abstract can be much longer than a paper abstract. I would consider splitting this into two paragraphs and setting up each takeaway a little more from the needs prior. I find the two lists a little disconnected.

Addressed: connected one-page abstract; first-person ownership retained. | 02-abstract.tex:7

A05 - 30 October 2026, 6:16 pm Pacific

I think you need something like this at the start of the abstract to help the reader understand where you're going with it.

Addressed: connected one-page abstract; first-person ownership retained. | 02-abstract.tex:7

I01 - 30 October 2026, 6:21 pm Pacific

For me this is missing a high level intro of why we should be interested in biomimetic humanoid robots

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:6

I02 - 30 October 2026, 6:24 pm Pacific

This paragraph is less a goal and more a motivation which I think should proceed the previous paragraph

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:6

I03 - 30 October 2026, 6:25 pm Pacific

Not sure this is a goal specifically and more motivation for what this work could also be beneficial for. You don't do anything related to this in your work

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:6

I04 - 30 October 2026, 6:26 pm Pacific

It's not clear to me what class of robot you mean here

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:6

I05 - 30 October 2026, 6:27 pm Pacific

Good

Retained: positive comment or adequate local explanation. | 10-introduction.tex:6

I06 - 30 October 2026, 6:37 pm Pacific

This doesn't belong in the motivation. Abstract or contributions

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:6

I07 - 30 October 2026, 6:41 pm Pacific

I think you can say this without the previous sentence. You're just trying to motivate the work here.

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:6

I08 - 30 October 2026, 6:42 pm Pacific

Good. Maybe say 'a robot knee' so it's a little more generalised

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:6

I09 - 30 October 2026, 6:48 pm Pacific

Muscle routes? Tendon routes?

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:6

I10 - 30 October 2026, 6:43 pm Pacific

You're citing a dissertation that isn't 'this dissertation' being your one

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:6

I11 - 30 October 2026, 6:44 pm Pacific

I would make it more clear that this is just the first problem

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:6

I12 - 30 October 2026, 6:45 pm Pacific

Again, not part of motivation

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:6

I13 - 30 October 2026, 6:46 pm Pacific

Good

Retained: positive comment or adequate local explanation. | 10-introduction.tex:6

I14 - 30 October 2026, 6:47 pm Pacific

I think this doesn't belong in the motivation either. Abstract or have a contributions section.

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:6

I15 - 30 October 2026, 6:48 pm Pacific

This is good but you could strengthen it by moving those contributions that are dotted in the motivation to here.

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:6

I16 - 30 October 2026, 6:50 pm Pacific

This is missing the 'why' of the work for me

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:22

I17 - 30 October 2026, 6:49 pm Pacific

Not sure where the trunk begins and ends.

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:22

I18 - 30 October 2026, 6:51 pm Pacific

I think this is the same as what you mean by routes previously but that could be clarified

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:22

I19 - 30 October 2026, 6:52 pm Pacific

This is one very long sentence and you lost me in what you're trying to say

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:22

I20 - 30 October 2026, 6:54 pm Pacific

It should be clear to me from the motivation paragraphs why you have chosen each of these objectives

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:28

I21 - 30 October 2026, 6:52 pm Pacific

Good

Retained: positive comment or adequate local explanation. | 10-introduction.tex:28

I22 - 30 October 2026, 6:55 pm Pacific

I'm not sure this is described earlier?

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:28

I23 - 30 October 2026, 7:02 pm Pacific

This only really gets one motivation sentence - I had to go back and check why you were doing this

Addressed in motivation, project chronology, or objectives. | 10-introduction.tex:28

I24 - 30 October 2026, 7:03 pm Pacific

This and the figure are good

Retained: positive comment or adequate local explanation. | 10-introduction.tex:28

S01 - 30 October 2026, 7:04 pm Pacific

I think you need to make it clear that this is your work in the first sentence

Addressed with targeted wording or retained explanation. | 50-database.tex:4

S02 - 30 October 2026, 7:05 pm Pacific

Is this someone who is building a model?

Addressed with targeted wording or retained explanation. | 50-database.tex:4

S03 - 30 October 2026, 7:07 pm Pacific

I think it would be helpful to the reader if you put 'clickable' items on the site in italics

Addressed with targeted wording or retained explanation. | 50-database.tex:4

S04 - 30 October 2026, 7:08 pm Pacific

Was?

Addressed with targeted wording or retained explanation. | 50-database.tex:23

S05 - 30 October 2026, 7:08 pm Pacific

What keywords?

Partial: verified monthly-watch terms added; historical manual search terms not recoverable. | 50-database.tex:27

S06 - 30 October 2026, 7:11 pm Pacific

I feel some kind of flow diagram for the order of operations could be helpful

Addressed through an explicit ordered workflow; no additional flowchart needed. | 50-database.tex:23

S07 - 30 October 2026, 7:16 pm Pacific

I think it would be really helpful to pick an example that you can walk through of how you'd use this

Addressed with targeted wording or retained explanation. | 50-database.tex:52

S08 - 30 October 2026, 7:12 pm Pacific

I'm assuming this is mentioned in the background section

Retained: positive comment or adequate local explanation. | 50-database.tex:86

S09 - 30 October 2026, 7:16 pm Pacific

This feels like more of an abstract for the subsection rather than the section itself. Where is the substance of what was done, examples of things not working etc?

Partial: personal speed/quality judgment stated; genuine AI-triage example remains pending. | 50-database.tex:71

S10 - 30 October 2026, 7:18 pm Pacific

Does the database directly tell you these differences or connect the papers? I think it's good to try to use the example but I'm not sure of the point.

Addressed with targeted wording or retained explanation. | 50-database.tex:86

S11 - 30 October 2026, 7:19 pm Pacific

I'm not sure how this paragraph relates to the database?

Addressed with targeted wording or retained explanation. | 50-database.tex:86

S11r - 30 October 2026, 7:20 pm Pacific

These paragraphs should probably start with the point you're trying to make and then show evidence

Addressed with targeted wording or retained explanation. | 50-database.tex:86

S12 - 30 October 2026, 7:21 pm Pacific

> Muscle pathway? Neural pathway?

Addressed with targeted wording or retained explanation. | 50-database.tex:86

S13 - 30 October 2026, 7:22 pm Pacific

> This should probably be in the motivation section? I'm not sure what relevance it has to the database

Addressed with targeted wording or retained explanation. | 50-database.tex:86

S14 - 30 October 2026, 7:23 pm Pacific

> Is the take home point supposed to be that you can use the database to select the optimal pathway to use for your robot? and then you go do the model development stuff?

Addressed with targeted wording or retained explanation. | 50-database.tex:86

N01 - 30 October, 9:07 pm Pacific

> It would be really helpful to me if you could include a high level overview of how this chapter is structured. I think it's: Motivation + Background (Matlab and it's problems); Problem identification; Methods for integration of each model? (Li-Deng, Gait2392, ...). But then I get confused between model implementation vs using tools or you're trying to implement the tools. Are you trying to get all the components to work together? Or just outline that they work in a valid way?

Chapter opener and roadmap now define the coupled-model objective and verification sequence. | 60-sns.tex:6

N02 - 30 October, 8:40 pm Pacific

> What controller what model? You should set this up so that it's clear this is the motivation and not the beginning of a methodology

Opener names Deng controller, Li mechanics, AnimatLab, and the motivating transfer problem. | 60-sns.tex:4

N03 - 30 October, 8:42 pm Pacific

> Who is the we vs I?

Morrow and I explicitly names collaborative additions and presentation; subsequent work remains first person. | 60-sns.tex:10

N04 - 30 October, 8:42 pm Pacific

> However?

Current opening no longer uses that transition; argument is stated directly. | 60-sns.tex:16

N05 - 30 October, 8:43 pm Pacific

> hyphen typos?

Reviewed current compound terms and TeX dashes; no blanket typographic substitution. | 60-sns.tex:16

N06 - 30 October, 8:45 pm Pacific

> I think you could tie up this motivation in a stronger manner. Is the dated nature of Animatlab the primary reason for the whole chapter? The wider context of why it's useful to be able to easily test these different models feels stronger to me...

Editor age removed as the rationale; efficient controller/mechanics experiments lead. | 60-sns.tex:16

N07 - 30 October, 8:47 pm Pacific

to be able to efficiently test different neural controllers ?

Stated explicitly at the close of the motivating model section. | 60-sns.tex:16

N08 - 30 October, 8:47 pm Pacific

Why not?

Explains why isolated neural and mechanical checks cannot establish closed-loop response: signal timing, magnitude, and sign matter. | 60-sns.tex:169

N09 - 30 October, 8:48 pm Pacific

This doesn't say how you coupled them?

Describes MN output to activation/force blocks and mechanical measurements through afferent encoders. | 60-sns.tex:169

N10 - 30 October, 8:50 pm Pacific

Simulation?

Uses coupled simulations and mechanical model explicitly. | 60-sns.tex:171

N11 - 30 October, 8:50 pm Pacific

So importing the model or reproducing a short trace is what you were doing in MATLAB?

States both component tasks and subsequent coupled simulations; retains longer divergence examples. | 60-sns.tex:171

N12 - 30 October, 8:52 pm Pacific

Good

Retain current material. | 60-sns.tex:90

N13 - 30 October, 8:53 pm Pacific

Cross implementation is what you're trying to achieve with your work?

Roadmap makes implementation agreement one verification step within coupled-model experiments. | 60-sns.tex:6

N14 - 30 October, 8:55 pm Pacific

Is 'common' a specific connectivity specification? If yes, I would capitalise, if no, state what it is?

Replaced common with same neuron definitions, synaptic connections, and parameter values. | 60-sns.tex:90